

Why Rotating Fluids Make Polygons

Paper IV: Planetary and Astrophysical Applications

Gordon Speagle

Abstract

Applying the constraint-intersection principle from Papers I–II, we predict the polygon wavenumber for Saturn ($N = 6$, Rossby-limited), Jupiter north ($N = 8$, Thomson-limited), Jupiter south ($N = 5$, packing-limited), and Neptune ($N \in \{3, 4, 5\}$, packing-dominated). Saturn’s 10% oblateness is not a correction but the mechanism: it flips the Thomson eigenvalue λ_3 from -0.12 (unstable on a mean sphere) to $+0.01$ (marginally stable), with a critical latitude $\varphi_{\text{crit}} \approx 72^\circ\text{N}$ below which the hexagon cannot exist. We derive a first-principles prediction chain from atmospheric parameters and resolve the Onsager–Arnold tension via a two-timescale argument.

1 The three-scale selection mechanism

Three scales cooperate: *macro* (Onsager inverse cascade concentrates vorticity into N clusters—a physical hypothesis, not derived), *meso* (the constraint-intersection principle selects which N), and *micro* (Arnold/Thomson stability arranges the clusters as a regular polygon). The Onsager hypothesis is supported by the Siegelman et al. [2022] simulations ($N = 8$ north, $N = 5$ south in 2D turbulence) but remains empirically untested for Jupiter’s actual 3D dynamics.

2 Wavenumber selection: the constraint-intersection principle

Proposition 2.1 (Constraint-intersection selection). *The observed polygon wavenumber N is determined by the intersection of three constraints. Two are fully derived from the 2D Laplacian:*

1. **Rossby stationarity** (derived; for jet meanders): the QGPV eigenvalue problem $(\nabla^2 + \beta \partial_y)\psi = 0$ selects the stationary wavenumber n^* .
2. **Thomson stability** (derived; for discrete cyclone rings): $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$ gives an upper bound $N \leq N_{\text{max}}(\kappa_0/\kappa)$; the energy monotonicity $H_{N+1} > H_N$ (Proposition 2.2) then selects $N = N_{\text{max}}$.

The third has a derived skeleton plus one empirical input:

3. **Geometric packing** (derived skeleton + empirical factor): domain admissibility of $G(\nabla^2)$ on the punctured domain gives $\sin(\pi/N) \geq \varepsilon/R$ where ε is the vortex core radius (derived). The effective exclusion radius $r_{\text{excl}} \approx 1.3\varepsilon$ includes a β -drift buffer zone from nonlinear vortex dynamics [Gavriel and Kaspi, 2021]; the factor 1.3 is observational, not derivable from the linear operator.

The selected wavenumber is

$$N_{\text{selected}} = \max\{N : N \leq N_{\text{Thomson}}, \sin(\pi/N) \geq r_{\text{excl}}/R, n \text{ Rossby-stationary}\}, \quad (2.1)$$

where inapplicable constraints are dropped (e.g. Rossby stationarity does not apply to discrete cyclone rings, and packing does not constrain jet meanders).

Remark 2.1 (Constraint-intersection principle). Each planetary system activates a different subset of the three constraints in (2.1):

System	N_{Thomson}	N_{Rossby}	N_{pack}	Binding
Saturn ($76^\circ N$)	7	6	—	Rossby
Jupiter north	8	—	> 8	Thomson
Jupiter south	8	—	5	Packing

The framework is unified: all three constraints derive from the 2D Laplacian ∇^2 and its modifications. The packing constraint is not an external geometric condition: it is the domain admissibility condition for the Green's function G_ε of ∇^2 on the punctured domain $\Omega_N(\varepsilon) = \mathbb{R}^2 \setminus \bigcup_{k=1}^N B(z_k, \varepsilon)$, where ε is the vortex core radius. This Green's function exists and defines a well-posed BVP if and only if the excluded discs are pairwise disjoint:

$$|z_j - z_k| \geq 2\varepsilon \iff \sin(\pi/N) \geq \varepsilon/R.$$

The domain-admissibility condition establishes $r_{\text{excl}} \geq \varepsilon$ (the core radius) from the Laplacian alone. The observed enlargement $r_{\text{excl}} \approx 1.3\varepsilon$ at Jupiter's south pole arises from the nonlinear β -drift balance of finite-amplitude vortices on a rotating sphere [Gavriel and Kaspi, 2021]; this factor is not derived from the linear operator $\nabla^2 + \beta \partial_y$ but is an empirical input from the full vortex dynamics.

The two Laplacian-derived facets and their observational supplement are:

1. Spectral positivity of $G(\nabla^2)$: Havelock eigenvalue $\lambda_m > 0 \rightarrow$ Thomson bound (derived);
2. Domain admissibility of $G_\varepsilon(\nabla^2)$: disc non-overlap $\sin(\pi/N) \geq \varepsilon/R \rightarrow$ packing bound with $r_{\text{excl}} = \varepsilon$ (derived); the enlargement $r_{\text{excl}} \approx 1.3\varepsilon$ at Jupiter south is observational;
3. Rossby stationarity of $(\nabla^2 + \beta \partial_y)\psi = 0$: stationary mode $n^* \rightarrow$ wave quantization (derived).

The finite-core regularisation G_ε does not create a blob-stable window for $N = 8$: the corrected constrained Hessian (with the full $\partial^2(-\frac{1}{2} \ln s)/\partial x_j^2 = (2\Delta x^2 - s)/s^2$, $s = d^2 + 2\varepsilon^2$, not the $O(\varepsilon^2)$ truncation) remains negative throughout the admissible domain $\varepsilon/R < \sin(\pi/8)$. The $N = 8$ selection therefore requires the central vortex ($\kappa_0/\kappa \geq \frac{1}{2}$); blob regularisation alone does not provide an alternative stabilisation mechanism.

For Saturn, n^* is determined by the Rayleigh–Kuo condition for stationary waves on a jet. All quantities are evaluated in the frame rotating with Saturn’s System III period—the frame co-rotating with the deep interior. In this frame the background wind at the hexagonal jet is $U_{\max} \approx 120$ m/s eastward. At the center of a Gaussian jet, the effective PV gradient is $\beta_{\text{eff}} = \beta + U_{\max}/\sigma_{\text{jet}}^2$, where the jet curvature term dominates bare β by a factor ~ 14 for $\sigma_{\text{jet}} = 2.5 \times 10^6$ m [Pedlosky, 1987, Vallis, 2017]. Two estimates of n^* are available:

- *Simple scaling* (independent of σ_{jet}): $n^* = R_{\text{hex}} \sqrt{\beta/U_{\max}} = 5.6$, which rounds to $n = 6$ for any σ_{jet} .
- *Leading-order with β_{eff}* : $n^* = R \cos \varphi \sqrt{\beta_{\text{eff}}/U_{\max}}$, which gives 5.4 at $\sigma_{\text{jet}} = 2.5 \times 10^6$ m but is sensitive to the jet width ($n^* = 6.4$ at $\sigma_{\text{jet}} = 2.1 \times 10^6$ m).

The full numerical Rayleigh–Kuo eigenvalue computation yields $n^* = 6$ [Sánchez-Lavega et al., 2014]; the simple scaling reproduces this robustly, while the β_{eff} formula is sensitive to σ_{jet} within its $\sim 20\%$ observational uncertainty. We adopt $\sigma_{\text{jet}} = 2.5 \times 10^6$ m throughout, consistent with the Cassini jet profile measurements; all conclusions are verified to hold over the range $\sigma_{\text{jet}} \in [2.0, 3.0] \times 10^6$ m. Thomson allows $N \leq 7$; both criteria are satisfied at $N = 6$.

For Jupiter’s north pole ($N = 8$): the cyclones are discrete vortex clusters, not a jet meander; Rossby stationarity gives $n^* \approx 0.3$ (not physically relevant). *Caveat*: the 2D point vortex model requires the cyclone roots to extend through a barotropic column much deeper than the ring radius. Juno gravity data constrains Jupiter’s differential rotation to $\sim 3,000$ km depth [Kaspi et al., 2018], roughly 4% of R_J —making Jupiter the least observationally supported case for the 2D assumption among our applications. The model is nevertheless applicable if the polar cyclones are confined to a deep weather layer ($H_{\text{eff}} \sim 3,000$ km, $R_d \gg R$) or to a shallow tropospheric layer with reduced gravity; both scenarios are consistent with the Juno data (see Section Paper II, §13 for quantitative limits).

Thomson stability provides an *upper bound*: $\kappa_{\text{crit}}(9) = 1 > \kappa_0/\kappa$, so $N = 9$ is unstable; $\kappa_{\text{crit}}(8) = \frac{1}{2} \leq \kappa_0/\kappa$, so $N = 8$ is stable [Adriani et al., 2018]. But $N = 7$ is also stable ($\kappa_{\text{crit}}(7) = 0$), so stability alone does not select $N = 8$ over $N = 7$. The selection requires one additional ingredient: a reason for the system to prefer higher-energy configurations among the stable set.

Proposition 2.2 (Energy monotonicity and maximum- N selection). *Let $\Gamma = N\kappa$ be the total ring circulation, $L = N\kappa R^2$ the angular impulse (so R is fixed when Γ and L are fixed), and $r = \kappa_0/\kappa$ held fixed as N varies (both the ring and central circulations scale proportionally). Denote by r_{core} the vortex core radius—the minimum separation at which the point-vortex model is valid; equivalently, r_{core} is the unit of length in which the logarithm is taken, so that $\ln R > 0$ iff $R > r_{\text{core}}$. Then*

$$H_{N+1} - H_N = \Gamma^2 \left[\frac{2r - 1}{2N(N+1)} \ln R + \frac{1}{2} \left(\frac{\ln N}{N} - \frac{\ln(N+1)}{N+1} \right) \right]. \quad (2.2)$$

For $r \geq \frac{1}{2}$ and $R > r_{\text{core}}$, both terms are non-negative (first: $2r - 1 \geq 0$ and $\ln R > 0$; second: $\ln N/N$ is strictly decreasing for $N > e \approx 2.7$), so $H_{N+1} > H_N$ for all $N \geq 3$. Therefore the N -gon ring energy is strictly monotone in N : the maximum- H configuration within the Thomson-stable set is the ring with the largest N :

$$N_{\max}(\kappa_0/\kappa) \equiv \max \{ N' : \kappa_{\text{crit}}(N') \leq \kappa_0/\kappa \}.$$

Any dynamics that preferentially populates higher-energy stable configurations—whether Onsager negative-temperature equilibrium ($\beta < 0$, where the Boltzmann weight $e^{-\beta H}$ increases with H), sequential vortex injection (adding cyclones until Thomson instability prevents $N+1$; this requires fixed r , see Remark 2.2), or merger of smaller vortex clusters—selects $N = N_{\max}$. The mathematical content of the proposition is the monotonicity $H_{N+1} > H_N$; the physical mechanism by which the system reaches $N_{\max}$ is a separate question (see the discussion below and Section 1).

The stability threshold $\kappa_{\text{crit}}(8) = \frac{1}{2}$ is exactly the value at which $H_{N+1} > H_N$ becomes unconditional (i.e. R -independent): at $r = \frac{1}{2}$, term 1 vanishes identically and term 2 remains strictly positive.

Proof. Use the Hamiltonian $H = -\kappa^2 \sum_{j < k} \ln |z_j - z_k| - \kappa_0 \kappa \sum_j \ln |z_j|$ with $\kappa = \Gamma/N$ and $\kappa_0 = r\kappa = r\Gamma/N$.

Step 1: Product formula. For the regular N -gon $z_j = Re^{2\pi i j/N}$, the chord length between vortices j and k is $|z_j - z_k| = 2R \sin(\pi |j - k|/N)$. By $\mathbb{Z}_N$ symmetry the pair sum equals $N/2$ times the sum from one vertex:

$$\sum_{j < k} \ln |z_j - z_k| = \frac{N}{2} \sum_{s=1}^{N-1} \ln(2R \sin(\pi s/N)) = \frac{N}{2} [\ln N + (N-1) \ln R].$$

Here we used $\sum_{s=1}^{N-1} \ln(2 \sin(\pi s/N)) = \ln N$, which follows from evaluating the polynomial identity $(z^N - 1)/(z - 1) = \prod_{k=1}^{N-1} (z - e^{2\pi i k/N})$ at $z = 1$: the product of moduli is $\prod_{s=1}^{N-1} |1 - e^{2\pi i s/N}| = \prod_{s=1}^{N-1} 2 \sin(\pi s/N) = N$.

Step 2: Central vortex term. Since $|z_j| = R$ for all j : $\sum_j \ln |z_j| = N \ln R$.

Step 3: Closed form for H_N . Substituting Steps 1–2 with $\kappa = \Gamma/N$, $\kappa_0 = r\Gamma/N$:

$$H_N = -\frac{\Gamma^2}{2N} \ln N - \Gamma^2 \left(\frac{N-1}{2N} + \frac{r}{N} \right) \ln R. \quad (2.3)$$

Step 4: Difference $H_{N+1} - H_N$. Writing H_{N+1} from (2.3) with $N \rightarrow N+1$ and subtracting:

$$H_{N+1} - H_N = -\frac{\Gamma^2}{2} \left[\frac{\ln(N+1)}{N+1} - \frac{\ln N}{N} \right] - \Gamma^2 \ln R \left[\frac{N/2 + r}{N+1} - \frac{(N-1)/2 + r}{N} \right].$$

The coefficient of $\ln R$ simplifies by partial fractions:

$$\frac{N/2 + r}{N+1} - \frac{(N-1)/2 + r}{N} = \frac{N^2 - (N-1)(N+1)}{2N(N+1)} - \frac{r}{N(N+1)} = \frac{1 - 2r}{2N(N+1)},$$

where we used $N^2 - (N-1)(N+1) = 1$. Negating and rearranging gives (2.2).

Step 5: Sign of each term. Term 2: $f(x) = \ln x/x$ satisfies $f'(x) = (1 - \ln x)/x^2 < 0$ for $x > e$, so $\ln N/N > \ln(N+1)/(N+1)$ for all $N \geq 3$.

Term 1: $2r - 1 \geq 0$ for $r \geq \frac{1}{2}$; and $\ln R > 0$ for $R > r_{\text{core}}$. Hence both terms are non-negative, and term 2 is strictly positive, giving $H_{N+1} > H_N$. $\square$

Remark 2.2 (Physical status of the fixed- r assumption). The assumption $r = \kappa_0/\kappa$ fixed as N varies means that as more cyclones appear in the ring (N increases), each individual ring vortex becomes weaker ($\kappa = \Gamma/N$) and the central vortex weakens proportionally ($\kappa_0 = r\Gamma/N$). This is non-trivial: the central polar cyclone is a distinct dynamical object from the ring cyclones and need not respond to changes in N .

The assumption is justified when all vortices — ring and central — are drawn from the same turbulent condensation process, so their strengths are set by a common large-scale energy budget rather than by independent forcings. In the Onsager picture, the negative-temperature state is reached by the inverse cascade from the same vorticity reservoir, giving a fixed partition of total circulation between center and ring. If instead κ_0 is held fixed while $\kappa = \Gamma/N$ varies (independent central vortex), then $r = N\kappa_0/\Gamma$ grows with N and the monotonicity of $H_{N+1} > H_N$ strengthens further, so the maximum- N selection remains qualitatively intact. The marginal case is $\kappa_0 = 0$ (no central vortex): then $r = 0 < \frac{1}{2}$ and Term 1 in (2.2) is negative, so $H_{N+1} > H_N$ requires R large enough that Term 2 dominates. The proposition’s $r \geq \frac{1}{2}$ condition is therefore specific to the regime in which the central vortex is sufficiently strong relative to the ring vortices.

The selection of $N = 8$ over $N = 7$ is sensitive to which assumption is made. Under fixed r : $H_{N+1} > H_N$ for all $N \geq 3$ (Proposition 2.2), so Onsager selects the Thomson-maximum $N_{\max}$. Under fixed κ_0 : r grows with N , which strengthens the monotonicity and gives the same $N_{\max}$. Under the intermediate assumption $\kappa_0 = \text{const}$ while Γ also varies: the result depends on how $\Gamma(N)$ scales, which is not observationally constrained. Thus the $N = 8$ selection is robust across the two limiting assumptions (fixed r and fixed κ_0) but relies on the Onsager framework providing the link between N , Γ , and κ_0 —a physical hypothesis, not a mathematical consequence.

Observationally, Jupiter’s central cyclone differs from the ring cyclones: it has a smaller diameter ($\sim 4,000$ km vs. 4,000–7,000 km; [Adriani et al. 2018](#)) and appears more axisymmetric. Whether it formed from the same inverse-cascade process as the ring cyclones, or was generated by a distinct polar convective mechanism, is not resolved by current Juno data.

For Jupiter north, $\kappa_0/\kappa \approx 1$ with roughly 30% observational uncertainty (Juno; [Adriani et al. 2018](#)), giving a conservative range $\kappa_0/\kappa \in [0.7, 1.3]$. Since $\kappa_{\text{crit}}(8) = \frac{1}{2}$ exactly, this entire range satisfies the stability condition for $N = 8$. However, $\kappa_{\text{crit}}(9) = 1$, which lies *inside* the uncertainty interval: if $\kappa_0/\kappa \geq 1$ (the upper half of the range), then $N = 9$ is also Thomson-stable and the Onsager argument would select $N_{\max} = 9$ instead. The conclusion $N_{\max} = 8$ therefore relies on $\kappa_0/\kappa < 1$, which is consistent with the central estimate but not guaranteed by the current observational precision. More precisely: $\kappa_0/\kappa \leq 1$ gives $N_{\max} = 8$ (observed); $\kappa_0/\kappa > 1$ gives $N_{\max} = 9$ (not observed). The observed $N = 8$ is thus evidence that $\kappa_0/\kappa \lesssim 1$, providing a theory-derived constraint on the central vortex strength that could be tested with improved Juno wind measurements.

For Jupiter’s south pole ($N = 5$): $\kappa_{\text{crit}}(N) = 0$ for all $N \leq 7$, so the Thomson bound alone allows up to $N = 7$ regardless of the central vortex. The observed $N = 5$ is selected by the geometric packing constraint (Proposition 2.1, item 3). South-pole cyclones are individually $\approx 2\times$ larger in linear size than north-pole cyclones ($r_{\text{core}} \approx 3,500$ km vs. 2,500 km; [Adriani et al. 2018](#)), and each cyclone carries a β -drift exclusion zone that extends $\approx 30\%$ beyond the core radius [[Gavriel and Kaspi, 2021](#)], giving $r_{\text{excl}} \approx 1.3 r_{\text{core}} \approx 4,550$ km. At the south-pole ring radius $R \approx 8,000$ km ($\sim 7^\circ$ from the pole):

$$\sin(\pi/5) = 0.588 \geq \frac{r_{\text{excl}}}{R} = \frac{4,550}{8,000} = 0.569, \quad \sin(\pi/6) = 0.500 < 0.569.$$

Thus $N_{\text{pack}} = 5$, which is tighter than the Thomson bound $N_{\text{Thomson}} = 8$: packing, not stability, is the binding constraint at Jupiter’s south pole. The $\approx 30\%$ excess of r_{excl} over the bare cyclone radius is physically the β -drift buffer zone identified by [Gavriel](#)

and Kaspi [2021]—cyclone repulsion on a rotating sphere creates an effective “personal space” larger than the vortex itself.

3 Saturn’s hexagon

3.1 Eigenvalue branch disconnection (Saturn persistence)

The preceding subsection selected Saturn’s wavenumber $n = 6$ via Rossby stationarity. A natural follow-up: can smooth perturbations of the jet destroy the stationary wave, converting the hexagonal meander into an evanescent (decaying) mode? The answer is no, by a WKB branch-disconnection argument.

The Rayleigh–Kuo equation for stationary ($c = 0$) mode n on a zonal jet $U(y)$ reads $\hat{\psi}'' + V(\rho)\hat{\psi} = 0$, where $V = (\beta - U'')/U - n^2/R^2$ is the effective potential in log-polar coordinates. At the jet center ($y = 0$) of a Gaussian jet:

$$V(0) = \frac{\beta + U_{\max}/\sigma^2}{U_{\max}} - \frac{n^2}{R^2} = \frac{\beta_{\text{eff}}}{U_{\max}} - \frac{n^2}{R^2}.$$

For Saturn ($U_{\max} = 120$ m/s, $\sigma = 2.5 \times 10^6$ m, $\beta = 1.46 \times 10^{-12}$ m⁻¹s⁻¹, $n = 6$, $R = R_S \cos 76^\circ = 1.31 \times 10^7$ m): $V(0) = 1.72 \times 10^{-13} - 2.10 \times 10^{-13} = -3.8 \times 10^{-14}$ m⁻² (negative). The sign of $V(0)$ depends on σ_{jet} : $V(0) < 0$ for $\sigma_{\text{jet}} \gtrsim 2.3 \times 10^6$ m, $V(0) > 0$ for narrower jets. At the adopted $\sigma_{\text{jet}} = 2.5 \times 10^6$ m, $V(0) < 0$ with $|V(0)|/V_{\text{threshold}} = 1.6$ (a 60% margin above the sign change); this margin grows to $3.4\times$ at $\sigma_{\text{jet}} = 3.0 \times 10^6$ m. The Cassini jet profile measurements constrain $\sigma_{\text{jet}} \in [2.3, 3.0] \times 10^6$ m, placing $V(0)$ robustly in the negative regime.

Since $V < 0$, the WKB exponent $s^2 = V < 0$ forces oscillatory solutions with $s = i\alpha_\rho$ ($\alpha_\rho \neq 0$). Writing $s = \sigma + i\alpha_\rho$: $\text{Im}(s^2) = 2\sigma\alpha_\rho = 0$. When $\alpha_\rho \neq 0$ (oscillatory region): $\sigma = 0$. The elliptic ($\sigma = 0$) and hyperbolic ($\alpha_\rho = 0$) branches are disconnected: smooth jet perturbations cannot move the solution from the oscillatory branch to the evanescent branch at leading WKB order. This is a structural stability result for the Saturn selection mechanism: the $n = 6$ stationary wave exists because $V(0) < 0$, and no continuous deformation of the jet profile can remove it without first passing through $V(0) = 0$ (which requires changing the sign of $\beta_{\text{eff}}/U_{\max} - n^2/R^2$, a finite perturbation). The hexagon is therefore structurally stable at leading WKB order: it cannot be removed by infinitesimal jet perturbations.

3.2 The oblateness correction (Saturn’s shape selects N=6)

The analysis above treats Saturn as a sphere. Saturn is an oblate spheroid with flattening $f = (a-b)/a = 0.098$ ($a = 60,268$ km equatorial, $b = 54,364$ km polar). The oblateness changes the *sign* of the Thomson stability eigenvalue for $N = 6$ —it is not a small correction but the mechanism that determines whether the hexagon can exist.

Curvature on the oblate spheroid. The Gaussian curvature at geodetic latitude φ is $K(\varphi) = 1/(M(\varphi)N(\varphi))$ where $M = a(1-e^2)/(1-e^2\sin^2\varphi)^{3/2}$ and $N = a/(1-e^2\sin^2\varphi)^{1/2}$ ($e^2 = 1 - b^2/a^2$). On an oblate body, K is *largest* at the equator and *smallest* at the pole—the pole is flatter. For Saturn: $K_{\text{eq}}/K_{\text{pole}} = a^4/b^4 = 1.51$. Since positive curvature destabilises (Paper I, Proposition 6.9), the pole is the *most stable* point on the spheroid—closer to the flat-plane limit where $N_{\text{crit}} = 7$.

The critical result. Substituting the local curvature $K(\varphi)$ into the spherical C_1 formula (a local-curvature approximation valid for $KR^2 \ll 1$):

Model	$\lambda_3(N=6)$	Stable?
Mean sphere ($R_{\text{mean}} = 58,300$ km)	-0.12	No
Oblate spheroid at 78°N	$+0.01$	Marginally yes

The oblateness flips λ_3 from -0.12 to $+0.01$. On a mean sphere, the hexagon is Thomson-unstable and *should not exist*; on the oblate spheroid, it is marginally stable.

Critical latitude. The latitude φ_{crit} at which $\lambda_3 = 0$ is determined by $C_1(K(\varphi_{\text{crit}}) \cdot R^2) = f(3, 6) = 9/2$. For $R_{\text{ring}} = 15,000$ km: $\varphi_{\text{crit}} \approx 72^\circ\text{N}$. Saturn’s hexagon at 78°N is 6° above this critical latitude—comfortably in the stable region but not far from the boundary.

Proposition 3.1 (Oblateness-stabilised hexagon). *On an oblate spheroid with flattening f , the binding eigenvalue $\lambda_3(N=6)$ at the pole exceeds its mean-sphere value by $\delta\lambda_3 \approx (N-1) \cdot 2\xi f(2-f)/(1+\xi)^2$ at leading order in f , where $\xi = K_{\text{mean}}R^2$. For Saturn ($f = 0.098$, $\xi \approx 0.07$): $\delta\lambda_3 \approx 0.13$, which reverses the sign of λ_3 from -0.12 to $+0.01$.*

Mode coupling at non-polar latitudes. At 78°N , the Gaussian curvature varies by $\delta K/K \approx 4.2\%$ across the hexagon ring (from the curvature gradient $dK/d\varphi$). This $\cos(2\theta)$ variation couples Fourier modes m and $m \pm 2$, with coupling amplitude $\sim (\delta K/K) \cdot C_1 \approx 0.19$. Since the stability margin is only $|\lambda_3| = 0.01$, the mode coupling *exceeds* the margin by a factor of 19.

By the topological protection of the Morse index (Paper I, Remark 6.5), this mode coupling cannot change the *net* spectral flow for $N = 6$ (which has all eigenvalues strictly positive at the pole). Individual eigenvalue trajectories are perturbed, but the net count of instabilities is zero—the hexagon remains stable in the topological sense, even though the perturbative stability margin is formally violated.

Neptune and Uranus: oblateness is negligible. Neptune ($f = 0.017$) and Uranus ($f = 0.023$) are far less oblate. The polar curvature correction $\delta K/K \approx -4.5\%$ (Neptune) and -6.8% (Uranus) shifts λ_3 by $\lesssim 0.01$ —negligible compared to the Thomson eigenvalue at these planets’ ring radii. The Neptune prediction $N \in \{3, 4, 5\}$ (Section 7) is robust to spheroidal corrections (`extensions/oblate_spheroid.py`).

4 Jupiter’s polar polygons

4.1 North pole: $N = 8$ from Thomson + central vortex

The north-pole octagon ($N = 8$) is Thomson-limited. On the flat plane, $N_{\text{crit}} = 7$; the octagon’s critical eigenvalue is $\lambda_4 = 7 - 8 = -1 < 0$ (unstable). A central vortex of strength κ_0 shifts this by $+2\kappa_0$; the critical ratio is $\kappa_{\text{crit}}(8) = |\lambda_4|/2 = 1/2$. For $N = 9$: $\kappa_{\text{crit}}(9) = 1$; for $N = 10$: $\kappa_{\text{crit}}(10) = 7/4$. Juno imagery shows a central cyclone of roughly equal strength to the ring cyclones ($\kappa_0/\kappa \approx 1$), which stabilises $N = 8$ but not $N = 9$. The energy monotonicity $H_{N+1} > H_N$ (Proposition 2.2) then selects $N = 8$ as the Onsager maximum.

The K_0 interaction constrains the weather-layer depth. At the Rossby deformation radius $R_d \approx 2,500$ km (matching the observed cyclone radius at $H_{\text{eff}} \approx 3,000$ km, $g' \approx 0.3 \text{ m s}^{-2}$), the ring-to-deformation ratio $R_{\text{ring}}/R_d \approx 8,500/2,500 \approx 3.4$. The K_0 stability analysis gives:

R/R_d	N_{crit}	Regime
0	7	Barotropic limit
0.8	8	Deep weather layer
1.0	8	$\leftarrow$ requires $R_d \gtrsim R_{\text{ring}}$
1.5	7	
2.0	6	Shallow weather layer

$N = 8$ requires $R/R_d \lesssim 1$, i.e. $R_d \gtrsim 8,500$ km. This constrains the effective weather-layer depth: $H_{\text{eff}} = f^2 R_d^2 / g' \gtrsim 3,000$ km, a testable prediction for Juno gravity measurements.

4.2 South pole: $N = 5$ from geometric packing

The south-pole pentagon ($N = 5$) is packing-limited. With $R_{\text{ring}} \approx 8,500$ km, cyclone radius $r_{\text{cyclone}} \approx 3,500$ km, and β -drift buffer ≈ 1.3 [Gavriel and Kaspi, 2021]: the effective exclusion radius $r_{\text{excl}} \approx 4,550$ km gives $r_{\text{excl}}/R_{\text{ring}} \approx 0.535$. Since $\sin(\pi/6) = 0.500 < 0.535$: a hexagonal ring does *not* fit—the cyclones would overlap. Since $\sin(\pi/5) = 0.588 > 0.535$: a pentagonal ring fits. Therefore $N_{\text{pack}} = 5$.

The selection is *tight*: it requires $R_{\text{ring}} < 2r_{\text{excl}} = 9,100$ km for $N = 6$ to be excluded. The observed $R_{\text{ring}} \approx 8,500$ km satisfies this with 7% margin. A testable prediction: if the south-pole ring radius were $> 9,100$ km, the framework would predict $N = 6$ instead of 5.

At the north pole: $r_{\text{excl}}/R \approx 0.38$, so $N_{\text{pack}} \geq 8$ —packing is not the binding constraint.

4.3 The north–south asymmetry in one parameter

The entire $N = 8$ vs. $N = 5$ asymmetry reduces to a single dimensionless ratio:

$$\chi = \frac{R_d^{(\text{south})}}{R_d^{(\text{north})}} \approx \frac{r_{\text{cyclone}}^{(\text{south})}}{r_{\text{cyclone}}^{(\text{north})}} \approx \frac{3,500}{2,500} = 1.40.$$

The constraint intersection requires $\chi \gtrsim 1.30$ for the south-pole packing bound to tighten from $N = 6$ to $N = 5$. The observed $\chi = 1.40$ matches to 8%. Deriving χ from atmospheric physics (insolation asymmetry, moist convection) remains open: radiative scaling gives $\chi \sim 1.07$ (insufficient), suggesting that deep convective dynamics set the pole-to-pole deformation-radius contrast.

5 Neptune and Uranus predictions

For Neptune ($R = 24,600$ km, $\Omega = 1.08 \times 10^{-4} \text{ rad s}^{-1}$, $\varepsilon = 0.017$), the packing bound dominates. The deformation radius R_d is poorly constrained (1,500–2,500 km from thermal wind estimates); the predicted N depends sensitively on it:

R_d (km)	$r_{\text{excl}}/R_{\text{ring}}$	N_{pack}	N_{pred}
1,500	0.39	7	≤ 7
2,000	0.52	5	5
2,500	0.65	4	4

using $R_{\text{ring}} \approx 5,000$ km (estimated from the polar vortex extent) and β -drift factor 1.3. The prediction is $N \in \{3, 4, 5\}$ —packing-dominated, like Jupiter’s south pole. Neptune’s oblateness ($\varepsilon = 0.017$) is negligible compared to Saturn’s ($\varepsilon = 0.098$); it shifts λ_{min} by < 0.002 and does not affect N_{crit} .

For Uranus (axial tilt 98°): the extreme tilt disrupts the Onsager condensation mechanism by preventing a persistent polar jet. The prediction is $N = 0$ (no polar polygon).

Falsifiability: a Neptune observation of $N \geq 6$ would violate the packing bound for any $R_d > 1,500$ km; a Uranus polar polygon would challenge the tilt-disruption hypothesis.

6 Why the hexagon is uniquely two-dimensional

The hexagon ($N = 6$) is marginally stable in $d = 2$ ($\lambda_{\text{min}} = +0.5$) but unstable for $d \geq 3$ ($\lambda_{\text{min}} = -0.018$ at $d = 3$). This follows from the generalised Casimir $h(m, N, \Delta)$ of Paper III, Proposition 10.1:

d	$\Delta = (d-2)/2$	N_{crit}	$\lambda_{\text{min}}(N=6)$	Hexagon
2	0	7	+0.50	Stable
3	1/2	5	−0.02	Unstable
4	1	5	−1.08	Unstable
5	3/2	5	−2.93	Unstable

Saturn’s hexagon requires the logarithmic interaction ($d = 2$, $\Delta = 0$), which is the Green’s function of the 2D Laplacian. In 3D, the Coulomb interaction ($1/r$) destabilises the hexagon. The logarithmic interaction survives in a quasi-2D atmosphere because the weather layer is thin ($H_{\text{eff}} \ll R_{\text{planet}}$) and the barotropic mode dominates: the effective dynamics is 2D to leading order, with 3D corrections entering at $O(H_{\text{eff}}/R)^2$. This explains why hexagonal features are observed only on gas giants with thin weather layers (Saturn) and not on terrestrial planets or in 3D laboratory flows.

7 Broader observational context and predictions

7.1 Polygon quality metrics and cross-planetary comparison

The two planetary polygon types require different quality metrics. For *discrete vortex rings* (Jupiter), the natural measure is the geometric deviation from a perfect N -gon:

$$\sigma_{\text{geom}} = \frac{1}{N} \sum_{k=1}^N \left| \ln |z_k / z_k^{\text{ideal}}| \right|, \quad (7.1)$$

where z_k^{ideal} are the best-fit regular N -gon vertices. This measures positional scatter of discrete cyclones; small σ_{geom} means the ring closely approximates the Thomson equilibrium.

For *continuous jet meanders* (Saturn), the vortex positions z_k are not defined. The appropriate measure is the fractional radial displacement of the jet center:

$$\varepsilon_{\text{obs}} = A_n / \bar{R}, \quad (7.2)$$

where A_n is the half-amplitude of the n -th Fourier mode of the jet-center radial position and $\bar{R}$ is the mean ring radius. For Saturn’s hexagon ($n = 6$, $\bar{R} \approx 13,200$ km), the observed $\pm 2^\circ$ latitude oscillation [Godfrey, 1988, Antuñano et al., 2015] gives $A_6 \approx 1,900$ km and $\varepsilon_{\text{obs}} \approx 0.14$ (range 0.11–0.18 across measurement epochs).

These two metrics are *not interchangeable*: applying σ_{geom} to Saturn (by treating the meander peaks as point positions) yields 0.50, an order of magnitude larger than Jupiter’s 0.05–0.06, but this comparison is physically meaningless—it conflates meander amplitude with positional scatter. Table 1 reports both metrics, using each where it is appropriate.

Table 1: Planetary polygon configurations. κ_0/κ is estimated from Juno imagery [Adriani et al., 2018] using cyclone area as a proxy ($\sim 30\%$ systematic uncertainty). “—” indicates the quantity is not applicable.

System	N	Quality metric	κ_0/κ	Selection mechanism
Saturn hexagon	6	$\varepsilon_{\text{obs}} = 0.14$	N/A (jet)	Rossby stationarity
Jupiter north	8	$\sigma_{\text{geom}} = 0.06$	~ 1	Thomson + Onsager
Jupiter south	5	$\sigma_{\text{geom}} = 0.05$	—	Packing (Prop. 2.1, item 3)

7.2 Broader observational context

Beyond the three planetary configurations, the theoretical stability boundary $\kappa_{\text{crit}}(N)$ from Proposition Paper II, Proposition 9.1 can be compared with laboratory and numerical observations:

Laboratory experiments. Jansson et al. [2006] and Barbosa Aguiar et al. [2010] observe polygons with $N = 2$ through 6 in rotating-tank experiments by varying the differential rotation rate. All observed N values fall in the $\kappa_{\text{crit}} = 0$ range ($N \leq 7$), consistent with the Thomson stability boundary. No laboratory experiment has produced a stable polygon with $N \geq 8$ without a central vortex—also consistent. (The laboratory mechanism is barotropic instability of a shear layer, not Thomson point vortex dynamics, so the agreement is structural rather than quantitative.)

Numerical simulations. Siegelman et al. [2022] produce $N = 8$ (north) and $N = 5$ (south) vortex crystals from 2D turbulence on polar caps, with a central polar cyclone emerging alongside the ring in both cases. The central cyclone provides the stabilization required by Proposition Paper II, Proposition 9.1 for $N = 8$. The Siegelman et al. [2022] simulations include continuous small-scale forcing (representing convective injection) and large-scale dissipation, yet the polygon crystal is the persistent stationary state: the system is not frozen into a one-time equilibrium but continuously re-attracted to it. Within the 2D barotropic framework, this is numerical evidence that the constrained energy minimum acts as a *dynamical attractor* under turbulent forcing. The caveat (Section 1) is that these are idealized 2D simulations, not a faithful model of Jupiter’s three-dimensional polar dynamics; the agreement supports the Onsager picture but does not constitute a proof of its applicability.

Uranus. Akins et al. [2023] detect a single polar cyclone at Uranus’s north pole—a single-vortex ($N = 1$) configuration, below the polygon threshold. No ring structure has been observed. Whether the absence of a Uranian polygon reflects insufficient vortex formation (macro-scale constraint) or insufficient observations (the pole has been poorly resolved) remains open.

Neptune. No polar polygon has been observed, and Voyager 2 imagery is insufficient to resolve ring structure. However, the framework makes a falsifiable prediction.

Regime: At 84°N , $\beta_N = 2\Omega_N \cos(84^\circ)/R_N \approx 9.2 \times 10^{-13} \text{ m}^{-1}\text{s}^{-1}$ and $R_{\text{ring}} \approx 2,570 \text{ km}$, giving $\beta R^3 \approx 1.6 \times 10^7 \text{ m}^2 \text{ s}^{-1}$. For Jovian-scaled cyclones (core radius $r_c \approx 240 \text{ km}$, $V_{\text{max}} \approx 80\text{--}100 \text{ m s}^{-1}$), $\kappa \approx 1.2\text{--}1.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$ and

$$\frac{\beta R^3}{\kappa} \approx 0.10\text{--}0.13 \ll 1.$$

Neptune is firmly vortex-dominated—more so than Jupiter ($\beta R^3/\kappa \approx 0.6$), because its smaller radius ($R_N \approx R_J/3$) reduces R_{ring}^3 by a factor of ~ 25 despite a slightly larger β . The Rossby stationarity mechanism does not apply; the relevant bridge is Thomson/Onsager.

Prediction: The first-principles chain (Section 8.4) narrows the Neptune prediction substantially. Neptune’s polar ring radius is $R_{\text{ring}} \approx 3,000 \text{ km}$ ($R_N \sin 7^\circ$), which is $2.9\times$ smaller than Jupiter’s. Its Coriolis parameter at 83° is $f_N = 2.15 \times 10^{-4} \text{ s}^{-1}$, only $0.62\times$ Jupiter’s. Consequently, the deformation radius $R_d = \sqrt{g'H_{\text{eff}}}/f$ is $1.6\times$ larger than Jupiter’s for the same atmospheric structure, and the packing ratio $\varepsilon/R = R_d/R_{\text{ring}}$ is $4.7\times$ tighter:

R_d	ε/R	N_{pack}	N (no CV)	N (with CV)
500 km (shallow)	0.17	18	7	8
1,000 km (moderate)	0.33	9	7	8
1,300 km	0.43	7	7	7
1,500 km (Jupiter-like)	0.50	5	5	5

Neptune’s strong internal heat flux ($F_{\text{int}} = 0.43 \text{ W m}^{-2}$, $2.6\times$ absorbed solar) suggests vigorous convection, favouring $R_d \gtrsim 1,000 \text{ km}$. In this regime **packing is the binding constraint**, and the prediction narrows to $N \in \{3, 4, 5\}$ —a triangle, square, or pentagon.

The physical reason is geometric: Neptune’s ring radius is too small for many cyclones of Jupiter-like size to fit. The $4.7\times$ tighter packing ratio (from the combination of smaller planet and weaker Coriolis) means that even moderate-size cyclones crowd the ring. This is a qualitative difference from Jupiter north, where Thomson (not packing) is the binding constraint.

This prediction is testable by a Neptune polar orbiter (e.g. Trident-class mission): imaging the polar region at $\sim 100 \text{ km}$ resolution would resolve individual cyclones and determine N directly.

All observations are consistent with the theoretical stability boundary: $N \leq 7$ is observed without a central vortex, $N = 8$ is observed only with one, and no $N \geq 9$ ring has been reported in any system.

8 Discussion and limitations

8.1 Why 2D and not 3D

The argument depends critically on two-dimensionality. The 2D Green’s function $G(r) = -\ln r/(2\pi)$ is the unique dilation-invariant pairwise interaction with $h' < 0$ (Remark Paper I, Remark 2.2; Corollary Paper I, Corollary 2.4 lifts this to the N -body level). In 3D, the Green’s function $G(r) = -1/(4\pi r)$ has $G'(r) = 1/(4\pi r^2) > 0$ —the sign rule is violated, the N -gon is not an energy extremum along the spiral, and the stability analysis gives different results. The present mechanism—sign rule plus energy-Casimir stability—does not operate in 3D; whether other mechanisms can produce persistent polygonal vortex structures in three dimensions is a separate question.

8.2 What is novel

Attribution. The Havelock eigenvalue $\lambda_m = (N-1) - m(N-m)/2$ and the stability boundary $N \leq 7$ are classical [Havelock, 1931]. Theorem Paper II, Theorem 8.2 bridges this to the Onsager literature, where the polygon is often described as an energy maximum: it is a constrained minimum, compatible with but distinct from the unconstrained Onsager picture (Corollary Paper II, Corollary 8.3).

What is new: (1) the constrained-minimum identification and its Lagrange-multiplier derivation; (2) the analytic sign rule (Paper I, Theorem 3.1); (3) the three-scale separation and two-timescale Onsager–Arnold resolution; (4) the constraint-intersection framework (Proposition 2.1); (5) the RG classification of the logarithm as IR fixed point (Proposition Paper I, Proposition 2.1); (6) the spherical and hyperbolic stability boundaries with exact thresholds; (7) the curvature-stability dichotomy and spectral flow (Propositions Paper I, Proposition 5.5–Paper I, Proposition 5.6); (8) the algebraic-integer classification $N \in \{8, 9, 11, 15, 23\}$, golden ratio terminal case, binary subdivision, Pell–conformal inversion equivalence, and continued fraction structure (Section Paper I, §6); (9) the blob bridge, K_0 central-vortex computation, and circulation disorder resilience; (10) the first-principles prediction chain reducing planetary N to atmospheric parameters.

Geometry selects placement. A recurring theme of the series is that stability depends not only on N and the curvature, but on *where* the polygon sits. On Saturn, the oblateness gradient makes the pole more stable than lower latitudes: the hexagon exists at 78°N but not below the critical latitude 72°N (Section 3.2), because the Gaussian curvature decreases toward the pole and the binding eigenvalue λ_3 flips sign at φ_{crit} . Paper I, §7 establishes the abstract counterpart: on an arithmetic surface $\Gamma \backslash \mathbf{H}^2$, the L^2 -Morse index at a palindromic threshold is *fractional*—61% of the Bolza surface is unstable while the automorphism-fixed centre remains stable. In both cases the mechanism is the same: a *symmetry-enhanced point* (the geographic pole on Saturn, the Aut-fixed centre on the Bolza surface) concentrates the stabilising curvature correction, making the vortex polygon viable at that location when it would be unstable generically. The oblateness of Saturn and the automorphism group of the Bolza surface are playing the same mathematical role—they select the placement.

Relation to prior work. Bouchet and Venaille [2012] and Chavanis and Sommeria [1996] classify self-organized vortex states via statistical mechanics; Aref et al. [2003] reviews vortex crystal stability. Our contribution adds the energy-curvature sign rule, the constrained-minimum clarification, and the planetary constraint-intersection framework—

none of which appear in these references. The $N = 7$ quartic coefficient $\alpha_0 = 45/14$ extends Kurakin and Yudovich [2002] (who proved $\alpha_0 > 0$ without the value) and complements Schmidt [2004] (who computed $9/28$ in Sylvester-normalized coordinates).

8.3 Limitations and scope

The mathematical results—the constrained energy minimum (Paper II, Theorem 8.2), the sign rule (Paper I, Theorem 3.1), and the stability table—are proven for discrete point vortices. They apply directly to Jupiter’s polar cyclone rings. For Saturn’s hexagon, which is a continuous jet meander, the relevant mechanism is Rossby wave stationarity of the QGPV equation, not the point vortex energy landscape. The continuous and discrete descriptions share the logarithmic Green’s function; the vortex blob analysis (Section Paper II, §12) establishes that the stability boundary is preserved under regularization and that the $N = 7$ marginal case is resolved.

The Saturn bridge (Section Paper II, §12.3) connects the continuous QGPV eigenvalue problem to the point vortex Hessian via critical-layer concentration. The key mechanism is $\mathbb{Z}_N$ symmetry: for any $\mathbb{Z}_N$ -symmetric cat’s-eye structure, the lobe circulations are equal and the centroids lie on the N -gon vertices, so the constrained Hessian has the same inertia as the Havelock Hessian (Remark Paper II, Remark 12.3). This is verified numerically for both smooth (Stuart vortex) and step-PV (Prandtl–Batchelor) cat’s-eye models. The braid vorticity fraction vanishes as $O(1-\eta)$ in the Stuart concentration limit (Remark Paper II, Remark 12.3), satisfying the conditions of Proposition Paper II, Proposition 12.2.

The quantitative bridge at Saturn’s actual parameters can be closed as follows. The relevant concentration parameter is the *physical* jet width divided by the hexagon radius: $\varepsilon/R = \sigma_{\text{jet}}/R_{\text{hex}} = 2.5 \times 10^6 / (5.1 \times 10^7) = 0.049$, not the log-polar width $\sigma_{\text{lp}} \approx 0.15$ sometimes quoted. With $d_{\min} = 2 \sin(\pi/6) = 1$, the concentration condition Paper II, eq. 12.5 requires $\varepsilon/d_{\min} < \delta(6) = 0.096$. The physical ratio $0.049 < 0.096$ satisfies this condition. The Weyl perturbation bound gives $\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq 54\varepsilon^2 = 0.13$, which is 26% of the spectral gap $\lambda_{\min}(6) = 1/2$ —absorbed with a $3.9\times$ margin. The actual $O(\varepsilon^2)$ blob correction is *stabilizing* ($c_6 = 13/12 > 0$, shifting $\lambda_{\min}$ upward by $+0.003$), and the separatrix vorticity sheet contributes $O(\varepsilon^4)$ to the eigenvalue (quadrupole coupling of a thin dipole layer), negligible compared to the gap. Saturn’s finite jet width therefore *strengthens*, not weakens, the $N = 6$ stability.

The Saturn hexagon is therefore supported by three arguments: (i) Rossby stationarity (Proposition 2.1) selects $n = 6$ independently of the energy landscape; (ii) geometric consistency with the spherical stability boundary (Section Paper I, §5.1); and (iii) the critical-layer bridge (Remark Paper II, Remark 12.3), which establishes Hessian inertia preservation for $\mathbb{Z}_N$ -symmetric cat’s-eye structures—now verified quantitatively at Saturn’s physical parameters ($\varepsilon/d_{\min} = 0.049 < \delta(6) = 0.096$, Weyl perturbation 26% of spectral gap). All three arguments converge on $N = 6$.

The Onsager negative-temperature hypothesis is a physical assumption whose applicability to Jupiter’s polar region is not directly tested here. It is the key step that distinguishes $N = 8$ from $N = 7$ in the selection argument (Proposition 2.2): without negative temperature, the framework identifies $N = 8$ as *stable* but does not uniquely select it over $N = 7$.

The Siegelman et al. [2022] simulations provide the most direct evidence available. Their 2D turbulence on polar caps develops an inverse energy cascade that drives kinetic

energy to the largest scales consistent with the polar geometry—the signature of a Onsager negative-temperature state. The polygon crystal is the maximum- H configuration within the Thomson-stable set: starting from random initial conditions, the simulations converge to the polygon ring with a central cyclone, which is the $N_{\max}$ configuration under Proposition 2.2 (they observe $N = 8$ north and $N = 5$ south, matching the Juno observations; the south-pole discrepancy from the Thomson/Onsager prediction is attributed to geometric packing as discussed above). This is indirect but consistent evidence that the planetary system is near negative temperature. A direct test would require measuring the energy spectrum at the polar scale in Juno or a dedicated mission.

Cassini and Juno data are digitized from published figures.

Dynamical maintenance. The constrained energy minimum (Theorem Paper II, Theorem 8.2) is a static result: it identifies the polygon as the lowest-energy arrangement on the conserved-quantity surface, and by Arnold’s energy-Casimir stability theorem [Arnold, 1969] this minimum is Lyapunov stable under the conservative 2D Euler dynamics—perturbations stay bounded. Whether this translates into an attractor under convective forcing depends on the ratio $\mathcal{R} = \tau_{\text{inject}}/\tau_{\text{restore}}$.

For Jupiter’s $N = 8$ ring (Juno parameters: $R_{\text{ring}} \approx 7,500$ km, $\kappa_0/\kappa \approx 1$; Adriani et al. 2018), we use the full-cyclone circulation $\kappa = 2\pi r_c V_{\max} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$ with $r_c \approx 3,200$ km (the full Juno-observed cyclone radius) and $V_{\max} \approx 80 \text{ m s}^{-1}$. This is the appropriate measure of total vortex strength for computing orbital periods and eigenfrequencies. It differs from the Rankine-core estimate $\kappa_{\text{Rank}} \approx 3.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$ (using only $r_c \approx 700$ km, the solid-body rotation region) that enters the regime number $\beta R^3/\kappa$ in Section 1: the regime criterion requires only a characteristic vorticity scale, for which the core is sufficient.

- *Restoring:* The ring orbital period is $T_{\text{orb}} = 8\pi^2 R^2/[\kappa(N-1)] \approx 4.6$ days. The most marginal mode ($m = 4$) has Lagrangian eigenvalue $\lambda_{\text{eff}} \approx 3$ (bare eigenvalue -1 , raised to $\approx +3$ by the central cyclone with $\kappa_0/\kappa = 1$). The oscillation e-folding time is $\tau_{\text{restore}} = (\sqrt{\lambda_{\text{eff}}} \Omega)^{-1} \approx 0.4$ days.
- *Injection:* The upper-bound convective power into one cyclone’s footprint is $P = F_{\text{int}} \pi r_c^2$, where $F_{\text{int}} \approx 7.5 \text{ W m}^{-2}$ is Jupiter’s interior heat flux [Guillot et al., 2004]; with $r_c = 700$ km, $P \approx 1.15 \times 10^{13} \text{ W}$. The tropospheric kinetic energy per cyclone is estimated as $E_K \approx \frac{1}{2} \rho V_{\max}^2 \pi r_c^2 H$ with $\rho \approx 0.2 \text{ kg m}^{-3}$, $H \approx 40$ km, $V_{\max} \approx 80 \text{ m s}^{-1}$: $E_K \approx 4 \times 10^{19} \text{ J}$, giving $\tau_{\text{inject}} = E_K/P \approx 40$ days.

The ratio $\mathcal{R} \approx 40/0.4 \approx 100$: the restoring force operates $\sim 100\times$ faster than the upper-bound convective injection. Including the inverse-cascade efficiency (typically $\lesssim 1\%$ of the heat flux reaches cyclone scales) raises $\mathcal{R}$ to order 10^4 . The Juno four-year observational baseline provides a direct empirical lower bound: $\mathcal{R} \gtrsim 4 \text{ yr}/0.4 \text{ days} \approx 3500$. The polygon is firmly in the restoring-dominated regime; convective forcing is a persistent perturbation that the energy minimum absorbs, not a mechanism that competes with the minimum for selection of the configuration.

Two-timescale resolution of the Onsager–Arnold tension. The constrained energy minimum (Lyapunov stability) and the Onsager negative-temperature selection (ergodic exploration of the energy landscape) appear to be in tension: if the polygon is dynamically stable, how does the system ergodically explore away from it? The resolution is that they operate *sequentially*, not simultaneously:

- **Formation phase** ($\tau_{\text{form}} \sim \text{weeks to months}$): The inverse cascade concentrates same-sign vorticity into N coherent clusters. During this phase the system is turbulent and ergodic; the Onsager negative-temperature argument determines *which* N is selected (N_{max} , the highest-energy Thomson-stable configuration). The polygon has not yet formed; the clusters are accumulating and arranging.
- **Persistence phase** ($\tau_{\text{persist}} \sim \text{years to decades}$): Once the N clusters reach the polygon arrangement, Arnold stability takes over. The polygon is a constrained energy minimum with $\mathcal{R} \approx 100$; perturbations are bounded and the system is *no longer ergodic*—it is trapped in the stability basin. The polygon persists because it is dynamically stable, not because of ongoing statistical equilibrium.

The Onsager argument selects N *before* the polygon forms; the Arnold argument maintains it *after*. The ratio $\mathcal{R} \approx 100$ quantifies the transition: the dynamical restoring timescale (0.4 days) is much shorter than the formation timescale (~ 40 days), so once the polygon assembles, it freezes into the energy minimum essentially irreversibly on observational timescales.

Moist convection as vorticity source. The leading mechanism proposed for maintaining Jupiter’s polar cyclones in the observational literature [O’Neill et al., 2015] is moist convective plumes injecting cyclonic vorticity preferentially at high latitudes. The geometric argument is that near the pole, the β -effect preferentially suppresses anticyclonic plumes (their vorticity anomaly is opposed by the planetary background), while cyclonic plumes are unimpeded; the result is a net positive-vorticity injection into the polar cap. This mechanism is quite different from—and more specific than—the Onsager inverse-cascade picture invoked here.

The two mechanisms are complementary rather than competing, operating at different scales and addressing different aspects of the problem:

- **O’Neill et al. generation:** moist convection injects cyclonic PV into the polar region, selecting *sign* (cyclones, not anticyclones). This is the vorticity *source*.
- **Onsager organization:** the injected cyclonic PV undergoes an inverse energy cascade, clustering into N coherent vortices. This is scale-blind to the source mechanism.
- **Thomson/energy landscape:** the N clusters arrange into a regular polygon. This is determined entirely by the point-vortex energy function.

The Onsager inverse cascade (and the present paper) requires only that same-sign vorticity be present in the polar region; it does not predict the sign. The O’Neill mechanism provides that sign selection. Without it, the Onsager picture would predict equal probability of cyclonic and anticyclonic polar clusters—which is not observed. A complete theory of Jupiter’s polar polygons therefore requires both mechanisms: O’Neill for the vorticity source and polarity, and the present Thomson/Onsager framework for the organization and stability of whatever is produced.

8.4 Toward first-principles prediction

The constraint-intersection framework (Proposition 2.1) predicts N given observational inputs about the vortices themselves (κ_0/κ , r_{cyclone} , R_{ring}). A stronger result would predict

N from *atmospheric* parameters that are independently measurable without observing the polygon structure. The deformation-radius chain partially achieves this reduction:

1. *Planetary parameters* (Ω, R, φ) give $f = 2\Omega \sin \varphi$ and $\beta = 2\Omega \cos \varphi / R$ (exact).
2. *Atmospheric structure* (g', H_{eff}) —the reduced gravity and effective weather-layer depth, measurable from temperature profiles and gravity data—give the deformation radius $R_d = \sqrt{g' H_{\text{eff}}} / f$ (derived), which sets the cyclone core size $r_{\text{core}} \sim R_d$.
3. *Domain admissibility* $\sin(\pi/N) \geq R_d / R_{\text{ring}}$ gives N_{pack} (derived from the Laplacian).
4. *Thomson + Onsager* gives $N_{\text{Thomson}} = 8$ for $\kappa_0 / \kappa \geq \frac{1}{2}$, which is the generic outcome of Onsager condensation when a central vortex is present (derived).
5. $N = \min(N_{\text{pack}}, N_{\text{Thomson}})$ (constraint intersection).

For Saturn, the Rossby path replaces steps 2–4: $n^* = R_{\text{hex}} \sqrt{\beta / U_{\text{max}}}$, where U_{max} is the jet speed (atmospheric observation, not a vortex observation).

At Jupiter, the north–south asymmetry ($N = 8$ vs. $N = 5$) reduces to a single parameter: $\chi = R_d^{(\text{south})} / R_d^{(\text{north})}$, the ratio of deformation radii at the two poles. With $R_d^{(\text{north})} \approx 2,500$ km (matching the observed north-pole cyclone radius at $H_{\text{eff}} \approx 3,000$ km, $g' \approx 0.3 \text{ m s}^{-2}$), the constraint intersection gives $N = 8$ (Thomson-limited). At the south pole, $\chi \approx 1.34$ gives $R_d^{(\text{south})} \approx 3,350$ km; with the β -drift exclusion factor ≈ 1.3 [Gavriel and Kaspi, 2021], the effective exclusion radius $r_{\text{excl}} \approx 4,350$ km yields $N_{\text{pack}} = 5$ (packing-limited). The required $\chi \approx 1.34$ matches the observed cyclone size ratio $r_{\text{cyclone}}^{(\text{south})} / r_{\text{cyclone}}^{(\text{north})} \approx 3,500 / 2,500 = 1.40$ to within 5%.

What remains is to derive χ itself from atmospheric physics. The south pole receives $\approx 22\%$ more insolation at perihelion (Jupiter’s orbital eccentricity $e = 0.049$), but radiative scaling gives only $\chi \sim 1.07$ —insufficient. A complete derivation of χ from moist convective dynamics remains an open problem; the framework reduces the prediction to this single parameter.

Three configurations from two planets is consistent with the proposed mechanism but not sufficient to establish universality. Uranus and Neptune observations would substantially strengthen the case. We use “consistent with a general mechanism” rather than “universal” to reflect the available evidence.

8.5 Falsifiability

The framework makes five testable predictions whose violation would require fundamental revision:

1. A *stable* $N \geq 9$ *polar polygon* on any gas giant would violate the Thomson bound ($\kappa_{\text{crit}}(9) = 1$, requiring the central vortex to exceed the ring vortex—an extreme configuration not produced by Onsager condensation).
2. *Saturn’s hexagon below $72^\circ N$ latitude* would violate the oblateness sign-flip ($\lambda_3 < 0$ below φ_{crit}).
3. A *hexagonal feature in a 3D-dominated regime* (N_{crit} drops from 7 to 5 at $d = 3$; the hexagon is uniquely two-dimensional).

4. *Neptune* $N \geq 6$ would violate the packing bound for any deformation radius $R_d > 1,500$ km.
5. *Jupiter* $N = 8$ in shallow-layer models ($R/R_d > 1.5$) would violate the K_0 suppression that requires a deep weather layer.

References

- A. Adriani, A. Mura, G. Orton, et al. Clusters of cyclones encircling Jupiter’s poles. *Nature*, 555:216–219, 2018.
- Alex B. Akins, Mark D. Hofstadter, Bryan J. Butler, A. James Friedson, Edward M. Molter, Robert J. Sault, and Imke de Pater. Evidence of a polar cyclone on Uranus from VLA observations. *Geophysical Research Letters*, 50:e2023GL102872, 2023.
- A. Antuñano, T. del Río-Gaztelurrutia, A. Sánchez-Lavega, and R. Hueso. Dynamics of Saturn’s polar regions. *Journal of Geophysical Research: Planets*, 120:155–176, 2015.
- H. Aref, P. K. Newton, M. A. Stremler, T. Tokieda, and D. L. Vainchtein. Vortex crystals. *Advances in Applied Mechanics*, 39:1–79, 2003.
- V. I. Arnold. On an a priori estimate in the theory of hydrodynamical stability. *American Mathematical Society Translations*, 79:267–269, 1969. Original Russian: Izv. Vyssh. Uchebn. Zaved. Mat. 54 (1966) 3–5.
- A. C. Barbosa Aguiar, P. L. Read, R. D. Wordsworth, T. Salter, and Y. Hiro Yamazaki. A laboratory model of Saturn’s north polar hexagon. *Icarus*, 206:755–763, 2010.
- Freddy Bouchet and Antoine Venaille. Statistical mechanics of two-dimensional and geophysical flows. *Physics Reports*, 515:227–295, 2012.
- P. H. Chavanis and J. Sommeria. Classification of self-organized vortices in two-dimensional turbulence: the case of a bounded domain. *Journal of Fluid Mechanics*, 314:267–297, 1996.
- Nimrod Gavriel and Yohai Kaspi. The number and location of Jupiter’s circumpolar cyclones explained by vorticity dynamics. *Nature Geoscience*, 14:559–563, 2021.
- D. A. Godfrey. A hexagonal feature around Saturn’s north pole. *Icarus*, 76:335–356, 1988.
- Tristan Guillot, David J. Stevenson, William B. Hubbard, and Didier Saumon. The interior of Jupiter. In F. Bagenal, T. E. Dowling, and W. B. McKinnon, editors, *Jupiter: The Planet, Satellites and Magnetosphere*, pages 35–57. Cambridge University Press, 2004.
- T. H. Havelock. The stability of motion of rectilinear vortices in ring formation. *Philosophical Magazine*, 11:617–633, 1931.
- Thomas R. N. Jansson, Martin P. Haspang, Kare H. Jensen, Pascal Hersen, and Tomas Bohr. Polygons on a rotating fluid surface. *Physical Review Letters*, 96:174502, 2006.

- Y. Kaspi, E. Galanti, W. B. Hubbard, D. J. Stevenson, S. J. Bolton, L. Iess, T. Guillot, A. Blöcker, J. Levin, P. Lauritzen, W. M. Folkner, R. Helled, A. P. Ingersoll, R. L. Kirk, M. J. Mariani, P. Tortora, S. Wahl, and M. Zannoni. Jupiter’s atmospheric jet streams extend thousands of kilometres deep. *Nature*, 555:223–226, 2018. doi: 10.1038/nature25793.
- Leonid G. Kurakin and Victor I. Yudovich. The nonlinear stability of stationary rotation of a regular vortex polygon. *Chaos*, 12:574–595, 2002.
- Morgan E. O’Neill, Kerry A. Emanuel, and Glenn R. Flierl. Polar vortex formation in giant-planet atmospheres due to moist convection. *Nature Geoscience*, 8:737–740, 2015. doi: 10.1038/ngeo2510.
- J. Pedlosky. *Geophysical Fluid Dynamics*. Springer-Verlag, 2nd edition, 1987.
- A. Sánchez-Lavega, T. del Río-Gaztelurrutia, R. Hueso, et al. The long-term steady motion of Saturn’s hexagon and the stability of its enclosed jet stream under seasonal changes. *Geophysical Research Letters*, 41:1425–1431, 2014.
- Dieter S. Schmidt. The stability of the Thomson heptagon. *Regul. Chaotic Dyn.*, 9: 519–528, 2004.
- Lia Siegelman, William R. Young, and Andrew P. Ingersoll. Polar vortex crystals: Emergence and structure. *Proceedings of the National Academy of Sciences*, 119(17): e2120486119, 2022.
- G. K. Vallis. *Atmospheric and Oceanic Fluid Dynamics*. Cambridge University Press, 2nd edition, 2017.